

A 24-VERTEX TREE REFUTING THE MARCHESSAULT–MYNHARDT UNIQUELY-RADIAL EQUALITY

ABSTRACT. Let T be the tree on 24 vertices obtained by chaining four copies of the spider $S(2, 2, 1)$. Its maximum split-set is unique and has $|M| = 3$, and each of its four radial subtrees is uniquely radial of radius exactly 2, so T satisfies the hypothesis of Problem 19 of Marchessault and Mynhardt (Problem 2 of [arXiv:2105.04611v2](#)). We exhibit a maximal boundary independent broadcast on T of cost 9, whence

$$\gamma_b(T) = 8 \leq i_{\text{bn}}(T) \leq 9, \quad \text{while} \quad \gamma_b(T) + \left\lceil \frac{|M|+1}{3} \right\rceil = 10.$$

Both admissible values differ from 10, so the answer to that problem is *no*. The published inequality $i_{\text{bn}}(T) \leq \gamma_b(T) + \lceil (|M| + 1)/3 \rceil$ is untouched. The phrase “uniquely radial” carries no numbered definition in the source; we use it in the sense fixed by its use there, as explained in the introduction.

1. INTRODUCTION

Throughout, T is a tree. A *broadcast* on T is a function $f: V(T) \rightarrow \{0, 1, \dots, \text{diam } T\}$ with $f(v) \leq e(v)$ for every v , where $e(v)$ is the eccentricity of v ; we write $V_f^+ = \{v : f(v) > 0\}$ and $\sigma(f) = \sum_v f(v)$. A vertex x *hears* $u \in V_f^+$ when $d(x, u) \leq f(u)$, and f is *dominating* when every vertex hears some broadcaster. The minimum cost of a dominating broadcast is the broadcast domination number $\gamma_b(T)$.

A broadcast is *boundary independent*, or *bn-independent*, when its broadcast balls overlap only in boundaries: with

$$N_f(u) = \{x : d(x, u) \leq f(u)\}, \quad B_f(u) = \{x : d(x, u) = f(u)\},$$

f is bn-independent when every vertex x that hears more than one broadcasting vertex satisfies $d(x, u) \geq f(u)$ for *all* $u \in V_f^+$. The *lower* boundary independent broadcast number $i_{\text{bn}}(T)$ is the *minimum* cost of a *maximal* bn-independent broadcast. In particular, exhibiting one maximal bn-independent broadcast of cost 9 proves $i_{\text{bn}}(T) \leq 9$; no minimality argument is needed.

Two further notions from [1] are needed. An edge xy is *covered* by $u \in V_f^+$ when $x, y \in N_f(u)$ and at least one of x, y is not in $B_f(u)$; U_f^E denotes the set of edges covered by no broadcaster. And for a diametrical path P of T ,

2020 *Mathematics Subject Classification*. 05C69, 05C05.

Key words and phrases. broadcast domination, boundary independent broadcast, lower boundary independence, split-set, radial subtree, uniquely radial, tree.

a set $M \subseteq E(P)$ is a *split-set* when every component-subpath of $P - M$ has even positive length and is diametrical in its own component of $T - M$ ([1], Definition 1). When M is a *maximum* split-set, the components of $T - M$ are the *radial subtrees* of T . As quoted in [1] (Theorem 1), a theorem of Herke and Mynhardt says that T is non-radial exactly when it has a split-set, and that then

$$(1) \quad \gamma_b(T) = \frac{\text{diam } T - |M|}{2}$$

for a maximum split-set M . A radial subtree is called *uniquely radial* in [1] when its broadcast domination number equals its radius and is attained by exactly one broadcast. The phrase carries no numbered definition there. What is defined is *radial*: [1] asks “which graphs G satisfy $\gamma_b(G) = \text{rad}(G)$? Such a graph G is called radial”. The force of “uniquely” is fixed by use, in the discussion of the tree T_1 of Figure 3 of [1], whose radial subtrees are declared not uniquely radial because each one “has a dominating broadcast of cost 2 from a central vertex, as well as the broadcast shown”. We adopt that reading, and it is the one clause of the hypothesis below that is a reading rather than a quotation.

Three results of [1] are quoted below and not reproved. First, the inequalities $\gamma_b(G) \leq i_{\text{bn}}(G) \leq \text{rad}(G)$ (the first from [1], Proposition 3, the second from the paragraph following it). Second, the upper bound ([1], Theorem 5)

$$(2) \quad i_{\text{bn}}(T) \leq \gamma_b(T) + \left\lceil \frac{|M| + 1}{3} \right\rceil,$$

which asserts “ $\leq$ ” only. Third, the theorem settling the two smallest cells ([1], Theorem 6): if the radial subtrees of T are uniquely radial and $|M| \in \{1, 2\}$, then $i_{\text{bn}}(T) = \gamma_b(T) + 1$. Since $\lceil (|M| + 1)/3 \rceil = 1$ for $|M| \in \{1, 2\}$, that theorem is exactly the equality case of (2) in those two cells. The problem we answer asks whether the equality persists beyond them.

Problem 1 ([1], Problem 19, p. 96, quoted verbatim; = Problem 2 of the e-print [2]). Let T be a tree whose radial subtrees are uniquely radial with radii at least 2 for any maximum split-set M . Is it true that $i_{\text{bn}}(T) = \gamma_b(T) + \lceil \frac{|M| + 1}{3} \rceil$?

The answer is no.

Theorem 2. *Let T be the tree of Section 2. Then T has a unique maximum split-set M , with $|M| = 3$; each of its four radial subtrees is uniquely radial of radius exactly 2; and*

$$\gamma_b(T) = 8, \quad 8 \leq i_{\text{bn}}(T) \leq 9, \quad \gamma_b(T) + \left\lceil \frac{|M| + 1}{3} \right\rceil = 10.$$

In particular $i_{\text{bn}}(T) \neq \gamma_b(T) + \lceil \frac{|M| + 1}{3} \rceil$, so the answer to Problem 1 is negative.

Theorem 2 contradicts no theorem of [1]. The inequality (2) asserts only “ $\leq$ ”, and $9 \leq 10$; the case $|M| \leq 2$ is untouched, because $|M| = 3$ here. What fails is only the conjectured equality, and it fails in the first cell the quoted theorem does not reach.

2. THE TREE

Let $S(2, 2, 1)$ be the 6-vertex spider with centre c , two legs ca_1a_2 and cb_1b_2 of length 2, and one pendant u at c ; it has $\text{diam} = 4$ and $\text{rad} = 2$. Let T be the chain of four copies of $S(2, 2, 1)$, obtained by adding the three edges $b_2^{(i)}a_2^{(i+1)}$ for $i = 1, 2, 3$. Thus T has a spine $v_0v_1 \cdots v_{19}$, reading

$$a_2^{(1)}a_1^{(1)}c^{(1)}b_1^{(1)}b_2^{(1)} \mid a_2^{(2)}a_1^{(2)}c^{(2)}b_1^{(2)}b_2^{(2)} \mid a_2^{(3)}a_1^{(3)}c^{(3)}b_1^{(3)}b_2^{(3)} \mid a_2^{(4)}a_1^{(4)}c^{(4)}b_1^{(4)}b_2^{(4)},$$

together with four pendants u_1, u_2, u_3, u_4 attached at the block centres $c^{(i)}$, which are the spine vertices v_2, v_7, v_{12}, v_{17} . Its 24 vertices are $v_0, \dots, v_{19}, u_1, \dots, u_4$ and its 23 edges are, in full,

$$(3) \quad \begin{aligned} &v_0v_1, v_1v_2, v_2v_3, v_3v_4, v_4v_5, v_5v_6, v_6v_7, v_7v_8, v_8v_9, v_9v_{10}, \\ &v_{10}v_{11}, v_{11}v_{12}, v_{12}v_{13}, v_{13}v_{14}, v_{14}v_{15}, v_{15}v_{16}, v_{16}v_{17}, v_{17}v_{18}, v_{18}v_{19}, \\ &v_2u_1, v_7u_2, v_{12}u_3, v_{17}u_4. \end{aligned}$$

Its leaves are $v_0, v_{19}, u_1, u_2, u_3, u_4$; the vertices v_2, v_7, v_{12}, v_{17} have degree 3 and every other spine vertex has degree 2. Finally $\text{diam } T = d(v_0, v_{19}) = 19$ and $\text{rad } T = 10$, attained at v_9 and v_{10} .

3. T SATISFIES THE HYPOTHESIS

Lemma 3. *(v_0, v_{19}) is the only pair of vertices of T at distance 19, so the diametrical path of T is unique and every split-set of T consists of spine edges.*

Proof. Every vertex of maximum eccentricity is a leaf, and the leaves are $v_0, v_{19}, u_1, \dots, u_4$. Now $d(v_0, v_{19}) = 19$, while $d(v_0, u_i) \in \{3, 8, 13, 18\}$, $d(u_i, v_{19}) \in \{18, 13, 8, 3\}$ and $d(u_i, u_j) \leq 17$ for $i \neq j$. $\square$

For $1 \leq p \leq 19$ write $e_p = v_{p-1}v_p$ for the p -th spine edge, so that by Lemma 3 a split-set is a set $\{e_{p_1}, \dots, e_{p_m}\}$ with $p_1 < \dots < p_m$. The component-subpaths then have lengths

$$(4) \quad p_1 - 1, \quad p_2 - p_1 - 1, \quad \dots, \quad p_m - p_{m-1} - 1, \quad 19 - p_m,$$

and these $m + 1$ lengths are even, positive and sum to $19 - m$.

Lemma 4. *For any split-set M of any tree T , $|M| \equiv \text{diam } T \pmod{2}$.*

Proof. The $|M| + 1$ component-subpath lengths are even and sum to $\text{diam } T - |M|$. $\square$

Lemma 5. *T has a unique maximum split-set, namely*

$$M = \{e_5, e_{10}, e_{15}\} = \{v_4v_5, v_9v_{10}, v_{14}v_{15}\},$$

and $|M| = 3$.

Proof. Forbidden positions. Cutting e_p makes both v_{p-1} and v_p endpoints of component-subpaths. Suppose some centre $c^{(i)}$ is an endpoint of a subpath of length L , and let y be that subpath's other endpoint. The pendant u_i lies in the same component of $T - M$ and satisfies $d(u_i, y) = L + 1 > L$, so the subpath is not diametrical in its component. As the centres are the spine vertices at positions 2, 7, 12, 17, this forbids $p \in \{2, 3, 7, 8, 12, 13, 17, 18\}$. Cutting e_1 or e_{19} leaves a subpath of length 0, which is not positive. The admissible positions are therefore

$$p \in \{4, 5, 6, 9, 10, 11, 14, 15, 16\}.$$

Parity. By (4), $p_1 - 1$ even forces p_1 odd; then $p_{j+1} - p_j - 1$ even forces $p_{j+1} \not\equiv p_j \pmod{2}$, so the parities alternate. Among the admissible positions the odd ones are $\{5, 9, 11, 15\}$ and the even ones are $\{4, 6, 10, 14, 16\}$. By Lemma 4, m is odd.

Spacing. Each subpath has even positive length, hence length at least 2, so $p_{j+1} \geq p_j + 3$.

Conclusion. If $m \geq 5$ then $p_m \geq p_1 + 3(m - 1) \geq 5 + 12 = 17$, and no admissible position is at least 17; with m odd this leaves $m \in \{1, 3\}$. For $m = 3$ we need p_1 odd, p_2 even and p_3 odd, all admissible, with gaps at least 3. If $p_1 = 5$ then $p_2 \in \{10, 14, 16\}$; $p_2 \in \{14, 16\}$ forces $p_3 \geq 17$, impossible, and $p_2 = 10$ forces $p_3 = 15$. If $p_1 = 9$ then $p_2 \in \{14, 16\}$ and $p_3 \geq 17$, impossible; if $p_1 = 11$ then $p_2 \geq 14$ and again $p_3 \geq 17$; and $p_1 = 15$ leaves no room. So $(p_1, p_2, p_3) = (5, 10, 15)$ is the only possibility. Its subpath lengths are 4, 4, 4, 4, each even and positive, and each subpath is diametrical in its component: the component of v_0 is $\{v_0, \dots, v_4, u_1\}$, in which $d(v_0, v_4) = 4$ while $d(u_1, v_0) = d(u_1, v_4) = 3$, so its diameter is 4; the other three components are identical copies. Hence $M = \{e_5, e_{10}, e_{15}\}$ is a split-set, and it is the unique one of maximum cardinality. $\square$

Because the maximum split-set is unique, the quantifier “for any maximum split-set M ” in Problem 1 ranges over one set only, and the value $|M| = 3$ entering the formula is unambiguous.

Lemma 6. *Each of the four radial subtrees of T is a copy of $S(2, 2, 1)$, and is uniquely radial of radius exactly 2.*

Proof. Each component of $T - M$ consists of one block's five spine vertices together with that block's pendant, i.e. is a copy of $S(2, 2, 1)$, with $\text{rad} = 2$ and unique centre c . We claim its only dominating broadcast of cost 2 is $\{c \mapsto 2\}$. A single vertex of strength 2 must be within distance 2 of all six vertices, and only c is: a_1 misses b_2 , u misses a_2 , and a_2 misses b_1 , each at distance 3. Two vertices of strength 1 cannot dominate, because a_2 forces a broadcaster in $\{a_1, a_2\}$, b_2 forces one in $\{b_1, b_2\}$ and u forces one in $\{u, c\}$, and these three requirements are pairwise disjoint. Hence $\gamma_b = 2 = \text{rad}$ and the γ_b -broadcast is unique. $\square$

Finally, T has a split-set, so T is non-radial and (1) applies:

$$(5) \quad \gamma_b(T) = \frac{\text{diam } T - |M|}{2} = \frac{19 - 3}{2} = 8,$$

which is corroborated from above by the dominating broadcast $\{v_2 \mapsto 2, v_7 \mapsto 2, v_{12} \mapsto 2, v_{17} \mapsto 2\}$ of cost 8 at the four block centres, and is consistent with $\gamma_b(T) = 8 < 10 = \text{rad } T$. Consequently Problem 1 demands

$$(6) \quad i_{\text{bn}}(T) \stackrel{?}{=} \gamma_b(T) + \left\lceil \frac{|M| + 1}{3} \right\rceil = 8 + \left\lceil \frac{4}{3} \right\rceil = 10.$$

4. THE CERTIFICATE

We use one maximality criterion of [1], quoted and not reproved.

Proposition 7 ([1], Proposition 7). *If f is bn-independent, $|V_f^+| \geq 2$, and every component of $T - U_f^E$ contains at least two broadcasting vertices, then f is maximal bn-independent.*

Lemma 8. *Let*

$$S = \{v_0, v_2, v_4, v_7, v_9, v_{12}, v_{14}, v_{17}, v_{19}\}$$

and let $f \equiv 1$ on S and $f \equiv 0$ elsewhere. Then f is a maximal bn-independent broadcast on T of cost 9. Consequently $i_{\text{bn}}(T) \leq 9$.

Proof. f is a broadcast, since $f \leq 1 \leq e(v)$ everywhere, and $\sigma(f) = |S| = 9$.

S is independent. Consecutive members of S are at spine distance 2, 2, 3, 2, 3, 2, 3, 2, never 1.

f is dominating. The nine members of S hear themselves. Of the remaining fifteen vertices: v_1 is heard by v_0 , v_3 by v_2 , v_5 by v_4 , v_6 by v_7 , v_8 by v_7 , v_{10} by v_9 , v_{11} by v_{12} , v_{13} by v_{12} , v_{15} by v_{14} , v_{16} by v_{17} and v_{18} by v_{17} ; and u_1, u_2, u_3, u_4 by v_2, v_7, v_{12}, v_{17} respectively, every block centre being a broadcaster.

f is bn-independent. The vertices hearing more than one broadcaster are exactly $v_1, v_3, v_8, v_{13}, v_{18}$: each is the midpoint of one of the five spine paths of length 2 joining two members of S , and no other vertex is within distance 1 of two members of S . Each of the five lies at distance exactly $1 = f(u)$ from both broadcasters it hears, and at distance at least 1 from every member of S , because none of them belongs to S (S being independent). So the required condition $d(x, u) \geq f(u) = 1$ for all $u \in V_f^+$ holds at each.

f is maximal, by Proposition 7. With $f \equiv 1$ on its support we have $N_f(u) = N[u]$ and $B_f(u) = N(u)$, so an edge is covered by u if and only if u is one of its endpoints. Hence U_f^E is the set of edges with neither endpoint in S , namely

$$U_f^E = \{v_5v_6, v_{10}v_{11}, v_{15}v_{16}\},$$

and $T - U_f^E$ has the four components

$$\{v_0, \dots, v_5, u_1\}, \quad \{v_6, \dots, v_{10}, u_2\}, \quad \{v_{11}, \dots, v_{15}, u_3\}, \quad \{v_{16}, \dots, v_{19}, u_4\},$$

carrying 3, 2, 2, 2 broadcasters respectively. Since $|V_f^+| = 9 \geq 2$, the criterion applies. $\square$

Proof of Theorem 2. Lemmas 5 and 6 put T inside the hypothesis of Problem 1, with $|M| = 3$, and (5) gives $\gamma_b(T) = 8$, so the demanded value is 10 by (6). Lemma 8 gives $i_{\text{bn}}(T) \leq 9$. For the lower bound, $\gamma_b(T) \leq i_{\text{bn}}(T)$ is quoted from [1], so $i_{\text{bn}}(T) \geq 8$. Hence $i_{\text{bn}}(T) \in \{8, 9\}$, and both values differ from 10. $\square$

SCOPE

Every step above is finite and recomputable from the edge list (3) and the broadcast printed there. The exact value of $i_{\text{bn}}(T)$ is *not* claimed as a result of this note: no exhaustive search over the broadcasts of T is carried out here, either by hand or by the verification program accompanying this note. Nothing in Theorem 2 depends on it, since $8 \neq 10$ and $9 \neq 10$ alike. No claim is made about any tree other than T , and no repair of Problem 1 is proposed: we neither propose a hypothesis restoring the equality nor characterise the trees for which it does hold.

A search of the boundary independent broadcast literature turned up no answer to Problem 1 and no tree with the properties above. The abstract page of [2] carries exactly two versions and no erratum; of the papers we could find citing [1], none answers this problem; and the later paper of Hoepner, MacGillivray and Mynhardt (doi:10.7151/dmgt.2546) answers neighbouring problems of the same list, but not this one. That search was not exhaustive, and it turned up no settled range of $|M|$ beyond the $|M| \in \{1, 2\}$ of the quoted theorem.

Exact verification. A verification program, `verify.py`, accompanies this note, together with the recorded output `verify.output.txt` of a run of it. It is Python 3.9+ and uses the standard library only, reads the edge list (3) and the broadcast f exactly as printed, and re-derives from the definitions restated in Section 1: that T is a tree of the stated order, size, degree sequence, diameter and radius; that its diametrical path is unique; the complete list of split-sets of T , hence $|M| = 3$ and the uniqueness of the maximum split-set; the radius, broadcast domination number and *number* of γ_b -broadcasts of each radial subtree, hence unique radially under the reading of Section 1; the value 8 that (1) demands of γ_b , together with the exhibited dominating broadcast of that cost; and that f is a legal broadcast of cost 9, is dominating, is bn-independent with exactly the stated set of multiply-heard vertices, has exactly the stated U_f^E , and satisfies the hypotheses of Proposition 7. The split-set enumerator prunes on the even-positive-length condition, and is therefore also checked against exhaustive enumeration of *all* edge subsets of the diametrical path on two small trees, where the two agree exactly.

Beyond these, the recorded run contains checks on further objects and quantities — a second broadcast on T , the domination number of T , a five-block chain, and small control trees — none of which is discussed in this note and none of which anything above relies on.

The program does not check the interpretive and bibliographic assertions of this note — the wording and numbering of Problem 1, the reading of “uniquely radial”, and the claim that no earlier work answers it — and it does not prove the results quoted from [1]: the formula (1), the inequality $\gamma_b \leq i_{bn}$ and Proposition 7 are transcribed as cited results, and only their arithmetic and their application to T are checked. The program prints one line per check and a closing verdict, and exits 0 only if every check passes; the recorded run reports 45 checks, all passing. Its own closing statement of what it does not cover is reproduced with it.

REFERENCES

- [1] E. M. Marchessault and C. M. Mynhardt, *Lower boundary independent broadcasts in trees*, Discuss. Math. Graph Theory **44** (2024), 75–99, doi:10.7151/dmgt.2434. Problem 19 is on p. 96, in Section 5 (Open Problems).
- [2] E. M. Marchessault and C. M. Mynhardt, *Lower boundary independent broadcasts in trees*, e-print arXiv:2105.04611v2 (27 June 2021). Problem 2 of its Section 5 (Open Problems) is Problem 19 of [1], clause for clause.